System Documentation

Electronics

PowerLine - Power Automation Unit PAU Engine
V4000R41
V4000Rx3
V4000Rx4
12V1600R50
Functional Description
Operating Instructions
Workshop Manual

E532829/02E

A Rolls-Royce
solution

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1 Safety

1.1 Important provisions for electronic products

General

This product may pose a risk of injury or damage in the following cases:
- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Assembly, maintenance, and repair without ESD („Electro Static Discharge“) equipment and ESD work station
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

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1.2 Intended use of electronic products

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:
- Within the permissible operating parameters in accordance with the (→ Technical data)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/Rolls-Royce Solutions contact/Service partner)
- In the original configuration of the delivery or in a configuration approved in writing by the manufacturer
- In compliance with all safety regulations and in adherence with all safety and warning notices in this manual
- Work which requires communication with the control units must only be carried out with a diagnosis software or with tools approved by the manufacturer.
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair

Any other use, particularly misuse, is considered as being contrary to the intended purpose. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

The specifications of the manufacturer will be amended or supplemented as necessary. Prior to operation, make sure that the latest version is used. The latest version can be found at:
- http://www.mtu-solutions.com

Modifications or conversions

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

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1.3 Personnel and organizational requirements

Organizational measures of the user/manufacturer

This manual must be issued to all personnel involved in operation, maintenance, repair, assembly, installation, or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair, assembly, installation, and transport personnel at all times.

Personnel must receive instruction on product handling and repair based on this manual. In particular, personnel must have read and understood the safety requirements and warnings before starting work.

This is important in the case of personnel who only occasionally perform work on or around the product. Such personnel must be instructed repeatedly.

Personnel requirements

All work on the product must be carried out by trained, instructed and qualified personnel only:
- Training at the Training Center of the manufacturer
- Qualified personnel from the areas mechanical engineering, plant construction, and electrical engineering and also for work with live parts

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair, assembly, installation, and transport in writing.

Personnel shall not report for duty under the influence of alcohol, drugs or strong medication.

Clothing and personal protective equipment

Always wear appropriate personal protective equipment, e.g. safety shoes, ear protectors, protective gloves, goggles, breathing mask. Follow the instructions concerning personal protective equipment in the respective descriptions of the individual activities or the manufacturer's documentation included in the delivery.

When working on live components, in particular on electrical switchgear or on high-voltage component, wear special protective clothing according to IEC 61482.

Safe handling of Substances of Very High Concern pursuant to the REACH regulation (Registration, Evaluation, Authorization and restriction of CHemicals): We recommend wearing protective gloves at all times in order to reduce risk when working.

TIM-ID: 0000040531 - 013

1.4 Safety regulations for commissioning and operation of electronic products

Safety regulations for commissioning

Install the product correctly and carry out acceptance in accordance with the manufacturer's specifications before putting the product into service. All necessary approvals must be granted by the relevant authorities and all requirements for initial startup must be fulfilled.

Whenever the product is subsequently taken into operation ensure that:
- All maintenance and repair work has been completed.
- All protective devices are in place.
- The power supply (power plug) must always be accessible to allow isolation of the equipment from supply voltage at any time.
- All components are properly grounded. Ground separately by means of a grounding stake as necessary.
- No persons wearing pacemakers or any other technical body aids are present.
- Adequate ventilation of the operating room must be ensured for any operating status.
- The product must be free of any damage, this applies in particular to lines and cabling.
- Protect battery terminals, generator terminals or cables against accidental contact.
- All connections are correctly arranged/assigned, e.g. polarization.

Immediately after putting the product into operation, make sure that all control and display instruments as well as the monitoring, signaling and alarm systems work properly.

Safety regulations during operation

The operator must be familiar with the control and display elements.

The operator must be familiar with the consequences of any operations performed.

During operation, the display instruments and monitoring units must be observed with regard to present operating status, violation of limit values and warning or alarm messages.

Malfunctions and emergency stop

Practice emergency procedures, especially emergency stopping, at regular intervals.

Take the following steps if any system malfunctions are detected or signaled by the system:
- Inform supervisor(s) in charge.
- Analyze the message.
- If required, carry out emergency operations e.g. isolate the equipment from the power supply.

Contact Service if the root cause of the malfunction cannot be clearly identified.

Operation of electrical equipment

When electrical equipment is in operation, certain components of these appliances are electrically live.

Follow the applicable operating and safety instructions when operating the devices and heed warnings at all times.

TIM-ID: 0000094294 - 001

1.5 Safety regulations for assembly, maintenance, and repair work on electronic products

Safety regulations for work prior to assembly, maintenance, and repair

Have assembly, maintenance, or repair work carried out by qualified and authorized personnel only.

Never carry out assembly, maintenance, or repair work with the product in operation, unless:
- It is expressly permitted to do so following a written procedure.

Attach "Do not operate" sign in the operating area or to control equipment.

Use the recommended special tools or suitable equivalents when instructed to do so.

Safety regulations when performing assembly, maintenance, and repair work

Special tools and lifting equipment

Use only proper and calibrated tools.

Removing, installing and cleanliness

Pay particular attention to cleanliness at all times.

Working on electrical and electronic assemblies

Always obtain the permission of the person in charge before commencing assembly, maintenance, or repair work and before switching off any part of the electronic system required to do so.

De-energize the relevant areas prior to working on assemblies.

Follow the safety rules:
- Disconnect the electric system from live parts.
- Reliably prevent inadvertent power-on.
- Verify absence of voltage using suitable measuring/test equipment e.g. voltage indicator.
- Connect conductors and grounding with a short-circuiting device, e.g. grounding bar.
- Cover adjacent live parts. Prevent access by unauthorized persons.

ESD (Electrostatic Discharge): Work on components which could be damaged by electrostatic discharge must always be carried out with appropriate equipment. Appropriate equipment are e.g. electrically conductive work surfaces or antistatic wristbands.

Do not damage cabling during removal work. When reconnecting, ensure that cabling cannot be damaged during operation by:
- Contact with sharp edges
- Chafing on components
- Contact with hot surfaces.

Do not secure cables on lines carrying fluids.

Do not use cable ties to secure lines.

Always use connector pliers to tighten union nuts on connectors.

Never use the product as a climbing aid.

Subject the device as well as the product to functional testing on completion of all repair work. The emergency stop function must be tested in particular. The emergency stop function test involves the deactivation of the engine controller power supply. This test may only be performed when the product is cold.

Store spare parts properly prior to replacement, i.e. protect them against moisture in particular. Package faulty electronic components or assemblies properly before dispatching for repair:
- Moisture-proof
- Shock-proof
- Wrapped in antistatic foil (as necessary)

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